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FACULTY OF
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INFORMATION TECHNOLOGY

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Assignment/Tutorial/Practical Report Cover Sheet
Assignment/Report Cover Sheet

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Subject Code:	TMF 2954

Project Title:	Java Programming - Project

Name of Lecturer:	Ts. Dr Lim Phei Chin

Due Date: 5 June 2026, 5.00 p.m	Date Submitted:	5 June 2026

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1. Unified Modelling language (UML) Diagram



Figure 1: UML Diagram

The MainApp creates the WelcomeScreen, which then creates the HomePage after the user enters their name. Users can access three main features Learn, QuizScreen and FlashCard through the HomePage.

The Learn class creates the SdgInformationScreen to display SDG 3 targets. The QuizScreen uses HealthQuestion (20 questions), GamificationEngine (stars and badges), GlobalTimerManager (60-second timer) and ScoreFileManager to save scores. The

ScoreFileManager writes data to scores.txt. The FlashCard class contains multiple Card objects, each containing a question and answer.The Learn class implements ILearningContent. The HealthQuestion class implements Evaluable. The GamificationEngine implements IGamification. The ScoreFileManager implements IDataHandler.

2. Classes and Interface Distribution

2.1 Learning Module

Creator: Alexandra Sharan Koleng (103187)		
Type	**Name**	**Description**
Class	WelcomeScreen	The WelcomeScreen constructor initialises and displays the landing interface for a mobile-styled Java Swing application themed around **SDG 3: Health and Well-Being**.
Class	HomePage	The HomePage class builds a personalized dashboard window sized like a mobile screen. It displays the user's name, a health graphic, a motivational quote, and a vertical list of three green menu buttons: "Learning Unit", "Quiz", and "Flash Cards." Clicking any of these buttons closes the homepage and opens the selected feature screen.
Class	Learn	Implements a desktop application designed as an interactive educational slideshow focused on **Sustainable Development Goal 3 (Good Health and Well-being)**
Class	SdgInformationScreen	A **Java Swing** graphical interface that displays a scrollable, mobile-optimised overview of the **UN Sustainable Development Goal 3 (Good Health and Well-Being) targets**.
Interface	ILearningContent	**Establishing the baseline capabilities** required for any educational slide variants in the application. It exposes the necessary method signatures to retrieve page titles, extract descriptive body text summaries, and resolve file paths for localised image assets.

Role: This integrated system acts as the core structural framework of the application, seamlessly handling entry-gate user registration and credentials authentication before passing session-aware profile states to a central navigation dashboard hub. It decouples the visual user interface engine from the underlying data model through strict object-oriented interface contracts, allowing the platform to dynamically feed scrollable, clip-protected global target indices alongside state-managed, paginated educational presentation viewports without compromising front-end system stability or backend code security standards.

How it contributes:

- WelcomeScreen
  - It serves as the initial landing gate, validating user credentials at startup to secure the runtime and personalize the experience.
- HomePage
  - It organizes the layout grid and presents clear button paths to access features like the learning module, quizzes, or profiles.

- Learn
  - It locks the interface into a mobile-style window boundary (410×730 pixels) to prevent visual clipping, offering users a choice between reading target lists or starting a slideshow.
- SdgInformationScreen
  - It bundles text and auto-scaled graphics into clean visual "cards" within a vertical scroll pane, ensuring high-density policy data is easy to read.
- ILearningContent
  - It standardizes how content objects retrieve text and image paths, decoupling raw slide data from the user interface code so new pages can be added easily without breaking the layout.

Interaction with other modules:

- WelcomeScreen
  - Upon successful login, it closes itself and passes the active username to initialize the main application dashboard.
- HomePage
  - It uses the username from the login screen to display a personalised greeting and closes itself to open whichever module the user clicks.
- Learn
  - It allows users to route back to the homepage and initialises the internal slideshow window to begin pulling slide data sequentially.
- SdgInformationScreen
  - It maintains session continuity using the active username and features a back button that disposes of the scroll feed to return the user to the main learning hub**.**
- ILearningContent
  - It acts as the data template for the content provider and allows the slideshow window to read and render slide data uniformly using polymorphism.

2.2 Quiz Module

Creator: Siti Nurafiqah Binti Azahari (106393)		
Type	**Name**	**Description**
Class	QuizScreen	An extended JFrame that handles the mobile view quiz flow. It displays active questions, dynamically toggles the option button based on question types.
Class	HealthQuestion	Safely encapsulates private fields for the question prompt, multiple-choice options array and correct answer string, providing validation via an equality method.
Interface	Evaluable	Defining behaviours required for any question variants in the application. It exposes the necessary method signature for retrieving prompts, matching options and validating user inputs.

Role: Delivering a dynamic, reliable and highly scalable self-assessment quiz engine for the SDG 3 application is the responsibility of the entire module cluster. It ensures that the quiz functions properly on a mobile dimension screen layout while maintaining strict code security standards by separating the presenting display from the underlying educational content.

How it contributes:

- Encapsulation and Data Security
  - *HealthQuestion* hides the sensitive question details behind private fields, ensuring that the data can only be accessed through public methods and cannot be accidentally corrupted by outer layout logic.
- Interface Inheritance
  - *HealthQuestion* utilizes interface inheritance by implementing Evaluable. This forces the class to inherit and implement the required method blueprints (*getQuestionPrompt()*, *getOptions()* and *checkAnswer()*), guaranteeing a predictable structure for the application.
- Interface Driven Polymorphism
  - The quiz engine can treat various question styles (MCQ or True/False) identically. This standardizes how data flows into the user interface.

Interaction with other modules:

- *HealthQuestion* implements *Evaluable,* inheriting its structural rules and filling its empty method container with active programming logic.
- *HealthQuestion* implements *Evaluable* interface. It relies on *Evaluable* as a structural guide to make sure it includes all the necessary methods for the quiz to work.
- *QuizScreen* receives the *username* from the previous screen to personalize the text. Once done with the quizzes, the user clicks the button, it closes itself and opens the *HomePage* or *LearnScreen* to return the user to the main menu.

2.3 Gamification Module

Creator: Sallen Liu Yee Ling (100925)		
Type	**Name**	**Description**
Class	GamificationEngine	Implements IGamification to award stars and badges based on quiz score percentage, manage a file-based top 10 leaderboard, and display results in a Swing GUI dialog. Uses method overloading for score evaluation.
Class	GlobalTimerManager	Manages a countdown timer for the quiz using a Swing Timer that ticks every second. Stops automatically when time runs out and triggers a callback function. Uses constructor overloading and implements ActionListener via actionPerformed.
Interface	IGamification	Defines the contract for the gamification system with three methods. Requires implementing classes to award stars, assign badges, and update the leaderboard. Implemented by GamificationEngine to provide the full reward logic.

Role: The Gamification Module serves as the engagement and motivation layer for the quiz application, transforming raw performance metrics into meaningful rewards and competitive elements through a Reward Distribution Engine (converts scores into 1-5 stars and badges), Time Pressure Manager (enforces deadlines with countdown and auto-submit), Competitive Tracking System (maintains persistent top 10 leaderboard), User Feedback Provider (delivers visual recognition via GUI dialogs), and Extensibility Layer (provides clean interface for module integration).

How it contributes:

- Increases User Motivation
  - Star rating (1-5) provide immediate visual feedback on performance level.
  - Achievements badges create collectible milestones (Master Scholar at 80%, Knowledge Seeker at 60%.
- Enhance Quiz Structure
  - *GlobalTimerManager* adds urgency and standardized time constraints.
  - Timer callback enables automatic quiz submission when time expires.
  - Reset functionality allows quiz retakes without restarting application.
- Ensure Data Persistence
  - Writes score to leaderboard.txt immediately after quiz completion.
  - Maintains top 10 scores sorted descending (hight first).
  - Survives application restarts through file-based storage.
- Supports Flexible Integration
  - Method overloading *(awardStart(score)* VS *awardStars(score,totalQuestion)*) accommodates different quiz lengths.
  - Interface – based design allows swapping gamification implementations.
  - Default parameter values (20 question) simplify usage.

Interaction with other modules:

- Main Application
  - Instantiates *GamificationEngine* (playerName) and *GlobalTimerManager* (timeLimit,callback), then coordinates the overall quiz flow.
- Quiz Engine Module
  - Calls *awardStarts(), awardBadges()*, and *updateLeaderboard()* to distribute rewards receives timer ticks and expiration callbacks from *GlobalTimerManager*.
- UI Module
  - Display countdown timer from *getTimeRemaining()*, renders stars and badges visually, and shows leaderboard dialog by display*LeaderboardGUI()*.
- File System
  - Persistently stores and retrieves leaderboard.txt containing top 10 player scores sorted in descending order

2.4 Integration Module

Creator: Emily Chiam Ee Wen (104000)		
Type	**Name**	**Description**
Class	ScoreFileManager	Manages user quiz scores by saving and loading results from the *scores.txt* file. Implements method overloading through multiple *saveScore()* methods and keeps track of the highest score for each user.
Class	FlashCard	Provides 10 interactive flashcards for revision. Users can reveal the answers and navigate between cards to strengthen their understanding of SDG 3 concepts.
Interface	IDataHandler	Defines the required methods for data storage operations, including saveScore(), loadScores(), getHighestScore(), displayAllScores(), and fileExists().

Role: The Integration Module is responsible for connecting all the different parts of the application together and providing additional features. It includes *ScoreFileManager* to save user scores to *scores.txt*, *FlashCard* to provide 10 review flashcards to help users prepare for the quiz and *MainApp* serves as the main entry point of the application.

How it contributes:

- Score Management
  - *ScoreFileManager* saves and loads user quiz scores from scores.txt. It keeps track of the highest score achieved by each user.
- Method Overloading
  - *ScoreFileManager* provides multiple *saveScore()* methods (with username/score, date and remark), allowing scores to be saved with different information such as date and remarks.
- Interface Implementation
  - *ScoreFileManager* implements *IDataHandler* interface, ensuring a consistent structure for score management operations.
- Exception Handling
  - *ScoreFileManager* uses try-catch blocks to handle file I/O exceptions, preventing the application from crashing when errors occur.
- Encapsulation
  - FlashCard contains an inner class Card, demonstrating encapsulation where the Card is only relevant within the FlashCard context.

Interaction with other modules:

- *QuizScreen* sends the user's final score to *ScoreFileManager* after the quiz is completed.
- *ScoreFileManager* stores scores in *scores.txt* and makes them available for future reference.
- *FlashCard* can be accessed from *HomePage* to help users revise before attempting the quiz.

3. Application Interface Screenshots

3.1 Welcome Screen



Figure 3.1: Welcome Screen

The Welcome Screen is the application's starting page. Users are required to enter their username before accessing the system. The username will be used to personalize the learning experience and record quiz scores. Input validation is implemented to ensure the username field is not empty.

3.2 Home Page



Figure 3.2: Home Page

The Home Page serves as the application's main navigation hub. Users can access the Learning Module, Quiz Module and Flash Card Module through the available buttons. The interface is designed with a mobile-friendly layout while running on a desktop environment.

3.3 Learning Module - SDG Targets

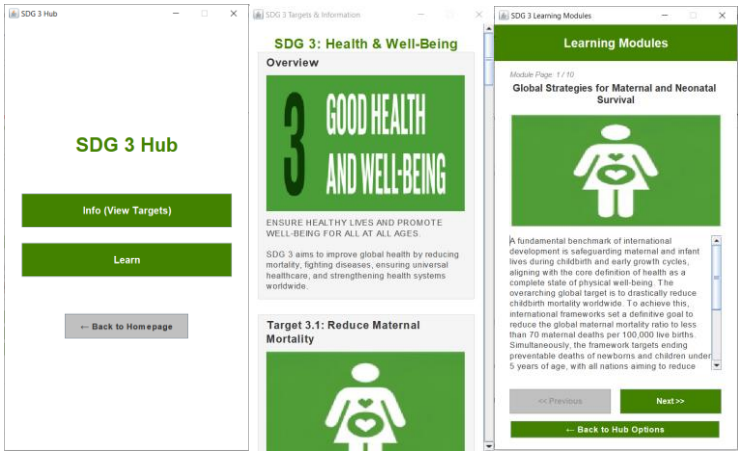


Figure 3.3: Learning Module

The Learning Module displays educational content related to Sustainable Development Goals (SDGs) 3 with descriptions and images. Users can scroll through all targets to learn about maternal health, communicable diseases, mental health, universal health coverage and more. Images are included to improve understanding and engagement.

3.4 Quiz Module

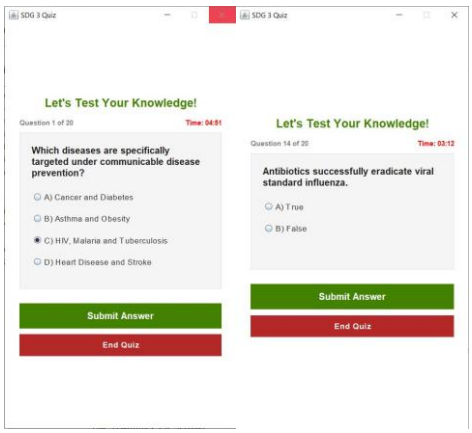


Figure 3.4: Quiz Module

The Quiz Module contains 20 questions including multiple choice and true/false questions, designed to assess the user's understanding of the SDG 3 topic. A countdown timer and a progress bar displaying the current question number are also included to increase user engagement and challenge.

3.5 Quiz Results with Gamification and Leaderboard

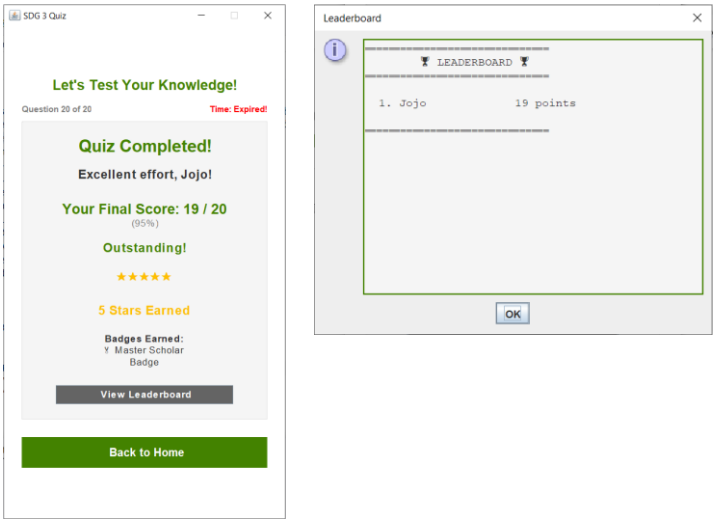


Figure 3.5: Quiz Results and Gamification

After completing the quiz, the application calculates the user's score and displays a motivational message based on performance. Gamification features include:

- Stars (1-5 stars based on percentage score)
- Badges (Master Scholar, Knowledge Seeker, Green Learner, Perseverance, Rainmaker)
- Leaderboard (users can click the "View Leaderboard" button to see top score)

These gamification features are used to encourage continuous learning and improve user motivation.

3.6 Flash Cards

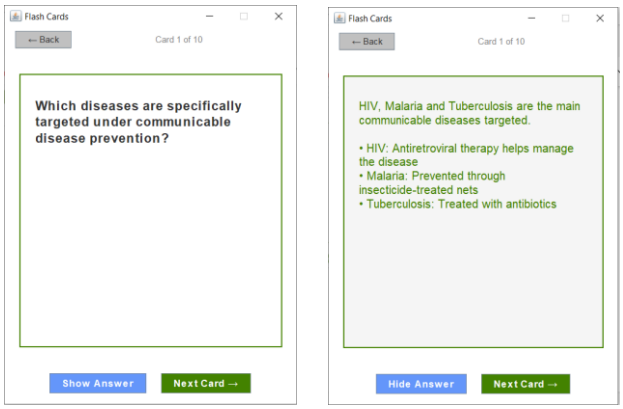


Figure 3.6: Flash Card Module

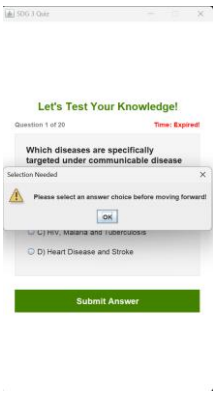
The Flash Card Module provides 10 review cards to help users review important SDG concepts. Each card displays a question on the front and the answer on the back. Users can click "Show Answer" to reveal the answer and "Next Card" to move to the next card. This module supports self-study and knowledge consolidation.

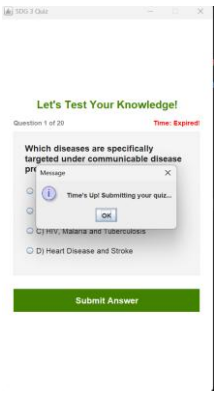
4. Testing and Debugging

4.1 Test Cases

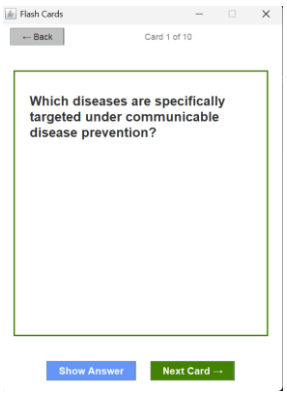
Test Case Name	Test By	Preconditions	Test Inputs	Expected Output	Actual Result	Pass/Fail
Welcome Screen Enter Name	Emily	App launched	Enter "Lin" , then Click Enter	Navigate to HomePage	HomePage opened successfully	Pass
HomePage Learn Button	Emily	HomePage displayed	Click Learn button	Open LearnScreen	LearnScreen opened successfully	Pass
Complete All Questions	Sallen	Quiz started	Answer all 20 questions	Show final score	Final score display successfully	Pass
Score 80 to 100 percent	Sallen	Quiz completed	Score 16-20/20	Show "Outstanding!"	“Outstanding!” display successfully	Pass
Score 0 to 19 percent	Alexandra	Quiz completed	Score 0-3/20	Show "Don't give up!"	"Don't give up!" displayed successfully	Pass
Timer Expired	Emily	Quiz started	Wait 60 seconds	Auto-submit quiz	When timer reached 0:00, quiz auto-submitted	Pass
Stars and Badges Display	Afiqah	Quiz completed	Score calculated	Show stars (1-5) and badges	Stars and badge well managed and perfectly displayed	Pass
Save Score to File	Afiqah	Quiz completed	Score calculated	Score written to scores.txt	Score successfully written in scores.txt	Pass
Flash Cards Open and Show Answer	Alexandra	HomePage displayed	Click Flash Cards, the click Show Answer	Open cards and display answer	Cards displayed the question and answer successfully	Pass

4.2 Debugging

<table>
<tr><td colspan="2">a) Problem: Learn Button Not Working</td></tr>
<tr><td>Problem Description: When clicking Learn button, the learning screen did not open.</td><td>Solution: Fixed ActionListener in HomePage.java to properly call new Learn(username)</td></tr>
<tr><td colspan="2">b) Problem: Timer Not Auto-Submitting Quiz</td></tr>
<tr><td>Before
 

Problem Description: When timer reached 0, the quiz did not automatically submit.</td><td>After
 

Solution: Changed condition from !timer.isTimeExpired() to timer.isTimeExpired()</td></tr>
<tr><td colspan="2">c) Problem: Score Not Saving</td></tr>
<tr><td>Problem Description: After quiz completion, scores were not being written to scores.txt.</td><td>Solution: Added ScoreFileManager.saveScore() call in showResults()</td></tr>
<tr><td colspan="2">d) Problem: Flash Card Question Not Displaying</td></tr>
<tr><td>Before
 

Problem Description: When opening Flash Cards, the question area showed blank.</td><td>After
 

Solution: Changed from JLabel to JTextArea with JScrollPane and setLineWrap(true)</td></tr>
</table>

5. Group Discussion Screenshots

